

What sort of cyberdeck do you want to create and why?

I am Deaf. I do not experience music through my ears. I experience it through my skin, my feet, my chest, the small bones of my hands resting on a speaker cabinet. For most of my life, that meant experiencing a lesser version of something designed for other people. My work is about changing that, not just for me, but for every Deaf and hard of hearing person who has been handed a lesser version and told to be grateful.

Sound travels as pressure waves. A computer can read those waves in real time, and translate them: a deep bass note becomes a slow rolling pulse against your palm, a sharp snare hit becomes a quick tap at your wrist, a crowd cheering becomes a warm wash across your whole body. I build the hardware that does this translation, and I deploy it at music festivals, in classrooms, on bicycles. Right now I am building a system that listens for the engine frequencies of approaching vehicles and vibrates powerfully when they get too close, because Deaf cyclists cannot hear a car coming up behind them, and that is a problem with a hardware solution.

To do this work in the field, at festivals, on roadsides, in school gyms with no reliable power, I needed a computer that did not depend on finding an outlet. So I am building one. The deck at the center of this residency is a portable computer that can run off almost any energy source: a solar panel left in the sun, a car battery in a parking lot, a handcrank, a thermoelectric plate resting against something warm. It finds power where power exists, and it keeps working. The chassis is shaped by where it goes: handles for carrying, a clip for wearing, a shell you can reprint at home when it breaks. No proprietary parts, no sending anything back to a manufacturer, no asking permission to fix your own tools.

The tutorial deliverable is the power system, packaged as an open circuit board you can order pre assembled for under \$100, with a guide Tracy and I are writing together. The guide is not a schematic. It is a story about claiming energy independence, written for someone who has never built anything before and wants to start. We are thinking of it as a zine that happens to have a parts list.

Tracy Held is my collaborator on this project, and she is the reason this work reaches people I cannot reach alone. As a screenwriter, community builder, and the grant writer whose work helped secure our active National Endowment for the Arts funding, Tracy has spent years learning how to bring unfamiliar ideas to people who were never supposed to be in the room. In Copenhagen she will lead the tutorial video, shape the documentation, and run the artist talk and community sessions with Dr. Cavdir's DHH community partners. She does not translate my work into plain language after the fact. She keeps it honest from the beginning.

That partnership with Dr. Doga Cavdir at ITU Copenhagen is the research spine of this visit. Together we have co authored two published papers on DHH music interfaces, presented at NIME 2024 and ACM SIGACCESS, with a third anticipated to grow from this residency. The work we are doing in Copenhagen directly seeds the next phase of our NEA funded DHH music

education research, connecting the hardware we build in SOFTER Lab to the students and communities it is ultimately designed for.

We are coming to Copenhagen to build something, to document it so anyone can build it too, and to share it with a community that understands, perhaps better than most, what it means to feel the world differently.